When Does Contiguity Help Watermark Detection? Sharp Phase Diagrams for Segment-Surviving Gumbel-max Watermarks

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Abstract

A Gumbel-max watermark is detected from one "pivot" per token, which is uniform on text the watermarked model did not produce and pushed toward 1 on text it did. After human editing, only a small fraction of tokens keep the watermark. The existing optimal theory treats these survivors as scattered independently, and its tests ignore token order. But, when a person keeps some AI sentences and rewrites the rest or pastes AI passages into their own writing, the survivors will form contiguous segments. We formalize this in our choice of alternative, which is that the watermark survives in segments of at least ℓ tokens, and we study the resulting problem within an explicit theoretical sandbox. We show that tests ignoring token order remain stuck at the old detection boundary however long the segments are, and that contiguity helps exactly when one segment carries enough evidence, as measured by its expected log-likelihood ratio. At the critical scale, Higher Criticism applied across blocks of tokens attains a sharp boundary, reaching full detectability exactly when one segment's expected log-likelihood ratio exceeds the log of the number of blocks $\log N$. A key-only count test captures most of the gain, and overall simulations agree with the theory.

1 Introduction

We continue the study of LLM watermarking from class, focusing on the Gumbel-max watermark (Aaronson, 2023). The model's sampling randomness is replaced by pseudorandom numbers derived from a secret key, so that the key holder can compute, for each token, a "pivot" that is uniform on human text and pushed toward 1 on watermarked text. Detection is formulated as a hypothesis test on these pivots, typically based on their sum or on Higher Criticism (HC), but it is made difficult when people edit the text. When a human revises watermarked text, only a fraction ϵ of the tokens keep the watermark, and detection becomes a problem of finding a sparse signal. Li, Ruan, Wang, Long and Su (2025b) model this scenario by treating each token as surviving independently, and they derive the detection boundary and show it is attained by an HC-type test.

However, editing rarely results in independent survival. A student might ask a model to rewrite an essay and then keep only a few of the AI sentences verbatim, or they might paste AI paragraphs into their own document, and these use cases result in text where the watermarked tokens form contiguous *segments* like whole sentences or passages. The tests in the i.i.d. theory, namely the sum test, HC and the truncated goodness-of-fit test of Li et al., depend only on the histogram of pivot values and not on where the large ones occur, so they cannot see this structure. Windowed detectors exist in practice, but there is no theory of when, and by how much, looking at token order helps. This paper asks the following question.

When the surviving watermark forms segments, how does the detection boundary change, and which tests attain it?

We keep the null hypothesis from class, namely that no token carries the watermark, and change only the alternative. Under our alternative the surviving tokens form runs of length at least ℓ that together cover at least an ϵ fraction of the text, in an arbitrary arrangement. The hardest arrangement turns out to be Li et al.'s model one level up, in which blocks of ℓ tokens (think of sentences) survive independently. The sparsity is then over blocks rather than tokens, which suggests summing the evidence within each block and applying HC across blocks.

Whether this helps depends on how much evidence a segment carries. A watermarked token is informative only to the extent that the model had freedom in choosing it. A forced choice has an exactly uniform pivot, and the more freedom the model had, the further the pivot is pushed toward 1. When the model is nearly deterministic, the evidence arrives in rare bursts. Only when the model departs from its top choice, which happens with probability about Δ_t, does the pivot land extremely close to 1, and we call such an occurrence a *watermark event*. The signal is then sparse at two levels, across tokens and within surviving tokens, and grouping tokens into a segment helps only if the segment accumulates enough evidence. Our results make this statement precise, with a token's expected evidence measured by $\mathrm{KL}(F_{P_t} \| \mathrm{Unif})$.

Main results. Within an explicit sandbox (Section 3), we prove four results.

(i) **Order-blind barrier** (Proposition 5.1). Any test that ignores token order has exactly the i.i.d. detection boundary, whatever the segment length.

(ii) **Coarse phase diagram** (Theorem 5.2, Figure 1). With $\epsilon = n^{-p}$, $\Delta = n^{-q}$ and segment length $\ell = n^{\gamma}$ ($\gamma \neq q$), detection is possible if and only if $2p + q < 1$ (the i.i.d. region) or $\gamma > q$ (segments containing many events).

(iii) **Sharp boundary at the critical scale** (Theorems D.1–5.3). When one segment's expected log-likelihood ratio is of order $\log n$, HC across blocks attains a sharp boundary. It lies strictly above the i.i.d. boundary and reaches full detectability exactly when a segment's expected log-likelihood ratio is at least $\log N$, the log of the number of blocks.

(iv) **A key-only test** (Corollary 5.4). Counting large pivots per block and applying HC across blocks needs only the key, has its own sharp boundary, and recovers most of the gain.

Simulations confirm these predictions, including outside the sandbox.

Some of the ingredients are known. Li et al.'s boundary $2p + q = 1$ already reflects that Δ acts like a sparsity parameter rather than a signal strength, and the general boundary we use for sparse mixtures (Theorem D.1) is equivalent to a result of Cai and Wu (2014), which we reprove in a form convenient for blocks of tokens. What is new is the analysis of *contiguously* surviving watermarks, namely the evidence-per-segment criterion, the proof that the i.i.d.-optimal order-blind tests are strictly suboptimal under contiguous survival, and sharp boundaries for both an oracle test and a key-only test.

The rest of the paper is organized as follows. Section 2 reviews the Gumbel-max watermark and Li et al.'s sparse-mixture model, Section 3 describes our sandbox, Section 4 quantifies the evidence carried by a token and by a segment (which drives our results), Section 5 presents the main results, Section 6 the simulations, and Section 7 their practical implications and limitations. We defer proofs to the appendices.

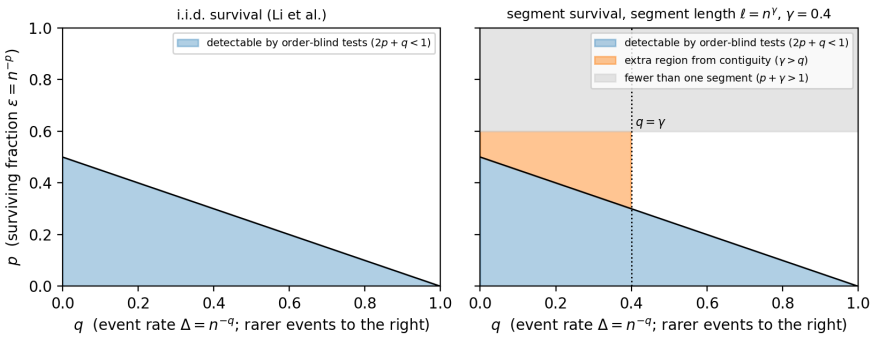


Figure 1: Coarse phase diagram (Theorem 5.2). Left: under i.i.d. survival, detection is possible iff $2p + q < 1$ (Li et al.). Right: under segment survival with segments of length n^{γ}, segment-aware tests also detect in the orange region, where segments are longer than $1/\Delta$ ($q < \gamma$). Order-blind tests remain confined to the blue region.

2 Background

2.1 The Gumbel-max watermark

At step t an LLM has a next-token distribution (NTP) $P_t = (P_{t,w})_{w\in\mathcal{W}}$ over the vocabulary $\mathcal{W}$. In the Gumbel-max watermark (Aaronson, 2023; Kuditipudi et al., 2024), a secret key and the recent context produce pseudorandom numbers $U_{t,w} \overset{\text{iid}}{\sim} \text{Unif}(0,1)$, one per vocabulary item, and the model emits

$$w_t = \operatorname*{argmax}_{w\in\mathcal{W}} U_{t,w}^{1/P_{t,w}} = \operatorname*{argmax}_{w\in\mathcal{W}} \big\{ \log P_{t,w} + G_{t,w} \big\}, \qquad G_{t,w} = -\log(-\log U_{t,w}).$$

The $G_{t,w}$ are standard Gumbel, so by the Gumbel-max trick w_t is an exact draw from P_t, and averaged over keys the watermarked text has exactly the model's distribution. The key holder recomputes the uniform number attached to the emitted token, which we call the *pivot* $Y_t = U_{t,w_t}$. Its law is as follows.

Lemma 2.1 (Pivot law; Li et al. 2025a)**.** *If token t was not produced by the watermarked sampler (e.g., a human wrote it), then $Y_t \sim \text{Unif}(0,1)$. If it was, then $Y_t \overset{d}{=} V^{P_{t,W}}$ with $W \sim P_t$ and $V \sim \text{Unif}(0,1)$ independent, and hence*

$$F_{P_t}(y) \equiv \mathbb{P}(Y_t \le y) = \sum_w P_{t,w}\, y^{1/P_{t,w}}, \qquad f_{P_t}(y) = \sum_w y^{1/P_{t,w}-1}, \qquad y \in [0,1].$$

The proof is in Appendix B. Because the key is independent of anything a human writes, a human-chosen token's pivot is exactly uniform, and the null hypothesis needs no model of human writing. A watermarked token chosen with certainty also has a uniform pivot, whereas a long shot ($P_{t,W}$ small) wins only when its key value was extreme, so its pivot is very close to 1.

The detectors from class use the scores $S_t = -\log(1-Y_t) \sim \text{Exp}(1)$ or the p-values $1 - Y_t \sim \text{Unif}(0,1)$ under the null, and reject for a large sum $\sum_t S_t$ or a large HC of the p-values. Both are *order-blind*, in the sense that they are unchanged if the tokens are shuffled. A key quantity throughout is the model's freedom $\Delta_t = 1 - \max_w P_{t,w}$, the probability that sampling departs from greedy decoding at step t. Let $\mathcal{P}_\Delta = \{P : \max_w P_w \le 1-\Delta\}$. Li et al. (2025a) show that the two-point law

$$P^\Delta = (1-\Delta,\ \Delta), \qquad f_\Delta(y) = y^\alpha + y^\beta, \quad \alpha = \tfrac{\Delta}{1-\Delta},\ \beta = \tfrac{1}{\Delta} - 1,$$

is least favorable in $\mathcal{P}_\Delta$ for their efficiency criterion when $\Delta \le 1/2$, and we use it as the canonical hard case.

2.2 Detection under edits as a sparse mixture

Li et al. (2025b) model editing by letting each token independently carry the watermark with probability ϵ_n, so that the pivots form a sparse mixture,

$$H_0 : \; Y_t \overset{\text{iid}}{\sim} \text{Unif}(0,1) \qquad \text{vs.} \qquad H_1 : \; Y_t \sim (1-\epsilon_n)\text{Unif}(0,1) + \epsilon_n F_{P_t}, \quad P_t \in \mathcal{P}_{\Delta_n}.$$

This is the rare/weak setting from class. With $\epsilon_n = n^{-p}$ and $\Delta_n = n^{-q}$, they show that detection is possible if $q + 2p < 1$ and impossible if $q + 2p > 1$, and that a truncated goodness-of-fit (Tr-GoF) test, a family of HC-like divergences between the empirical CDF of $1 - Y_t$ and the uniform CDF, attains the boundary adaptively. Sum-based tests, by contrast, reach only $q + p < 1/2$.

A simple count explains this boundary. The number of pivots above $1 - \Delta$ is $n\Delta \pm \sqrt{n\Delta}$ under the null, and the watermark adds about $n\epsilon\Delta$ (Lemma B.1). Detection therefore requires $n\epsilon\Delta \gg \sqrt{n\Delta}$, that is, $n\epsilon^2\Delta \to \infty$, which is the condition $2p + q < 1$. Much of our analysis repeats this calculation at the level of blocks.

Related work. Scan statistics and windowed or segmentation-based watermark detectors (Kirchenbauer et al., 2024; Pan et al., 2025; Li, Liu and Li, 2025) find watermarked spans, but none gives detection boundaries for contiguously surviving watermarks or compares them with the i.i.d.-optimal order-blind tests. Our proofs use the tools behind the "undetectable" region, which we stated but did not prove in class (specifically, in Lecture 3). Appendix A discusses related work in more detail and reviews these tools.

3 Problem formulation and sandbox

We now describe the idealized model in which all of our results are proved, and for each assumption we explain what it means, where it comes from, and what it leaves out. The final (edited) text has n tokens with NTPs $P_1, \ldots, P_n$, and the detector, holding the key, computes the pivots $Y_1, \ldots, Y_n$. Let $S \subseteq [n]$ be the unknown set of positions that still carry the watermark.

Assumption A1 (Pivot model; from Li et al.)**.** Given S and the NTPs, the pivots are independent with $Y_t \sim F_{P_t}$ for $t \in S$ and $Y_t \sim \text{Unif}(0,1)$ for $t \notin S$. Write $\mathbb{P}_S$ for their joint law.

Assumption A1 is the standard idealization of the statistical watermarking literature, with NTPs fixed and keys fresh at every step. It ignores that P_t depends on the edited context and that keys repeat when contexts repeat. The new ingredient of this paper is the alternative hypothesis.

Assumption A2 (Segment survival; new)**.** We test $H_0 : S = \emptyset$ against $H_1 : S \in \mathcal{S}_n(\epsilon, \ell)$, where

$$\mathcal{S}_n(\epsilon,\ell) = \Big\{ S = \textstyle\bigcup_j I_j : \; I_j \text{ disjoint intervals}, \; |I_j| \ge \ell, \; \sum_j |I_j| \ge \epsilon n \Big\}.$$

Assumption A2 requires the watermark to survive on at least ϵn tokens, in runs of at least ℓ tokens. The number, positions and lengths of the runs are unspecified, so tests must work for every arrangement and need no model of the editor. The single parameter ℓ controls contiguity. Taking $\ell = 1$ allows any set, which is the i.i.d. setting, while larger ℓ rules out fragmented survival. Such runs arise naturally from span-level editing and copy-paste. With context hashing, a token keeps

the watermark only if it and its h predecessors are unedited, so rewriting whole sentences leaves the kept sentences as runs, and a pasted passage of length ℓ_0 yields a run of length $\ell_0 - h$. Scattered token edits, by contrast, do *not* create runs, since if each token survives independently with small probability, runs have mean length about 1. Our lower bounds show that the hardest member of the class is *block-i.i.d. survival*, in which the text is tiled into n/ℓ blocks ("sentences") that are kept independently, which is Li et al.'s model one level up. The class ignores partial survival inside a segment and any distribution over segment lengths.

The remaining two assumptions are inherited from Li et al. and from the calibration used in class.

Assumption A3 (NTP regularity; from Li et al.)**.** For $t \in S$, $P_t \in \mathcal{P}_\Delta$, and the coarse results hold uniformly over this class. The sharp results instead fix the NTPs (Assumption A5).

Assumption A4 (Asymptotic calibration; as in class)**.** $\epsilon_n = n^{-p}$ with $p \in (0,1)$; write $L = \log n$. In the *coarse regime*, $\Delta_n = n^{-q}$ with $q \in (0,1)$ and $\ell_n = n^\gamma$ with $p + \gamma < 1$. In the *critical regime*, one segment's expected log-likelihood ratio is of order $\log n$, which happens either when $\Delta_n = n^{-q}$ and $\ell_n \Delta_n = \kappa \log n$, or when the NTPs are fixed and $\ell_n = c \log n$. There are then $N = \lfloor n/\ell_n \rfloor = n^{a+o(1)}$ blocks, with $a = 1 - q$ or $a = 1$ respectively, and we take $p < a$ (many segments). In these units Li et al.'s boundary $2p + q = 1$ reads $p = a/2$.

Finally, a test ψ may be any function of the pivots, and we judge it by its worst-case risk $R_n(\psi) = \mathbb{P}_\emptyset(\psi = 1) + \sup_{H_1} \mathbb{P}_S(\psi = 0)$, where the supremum is over all S and NTPs allowed by the assumptions. The problem is *detectable* if some test has $R_n \to 0$ and *undetectable* if every test has $R_n \to 1$. We distinguish *oracle* tests, which may use the NTPs, from *key-only* tests, which use only the pivots. Table 1 in Appendix A summarizes the sandbox.

4 Evidence per token and per segment

All of our results are driven by a single quantity, the amount of evidence that a watermarked segment carries. Because the pivots are independent, this evidence is a sum over tokens, and it is convenient to describe it through the moment generating function of a token's log-likelihood ratio. For an NTP P, let $m_P(\theta) = \mathbb{E}_{\mathrm{Unif}}[f_P(Y)^\theta]$ be the moment generating function of $\log f_P(Y)$ under the null (for a non-degenerate NTP with finite support, it is finite exactly for $\theta > -P_{\max}/(1 - P_{\max})$). Three of its values have a direct interpretation,

$$\log m_P(0) = \log m_P(1) = 0, \qquad (\log m_P)'(1) = \mathrm{KL}(F_P \| \mathrm{Unif}), \qquad \log m_P(2) = \log\big(1 + \chi^2(F_P \| \mathrm{Unif})\big).$$

The KL is the expected evidence of one watermarked token, and χ^2 is its second-moment information. A block of watermarked tokens therefore has null cumulant $\sum_t \log m_{P_t}(\theta)$ and expected log-likelihood ratio $\sum_t \mathrm{KL}(F_{P_t} \| \mathrm{Unif})$, so that *evidence adds over tokens*.

When the model is nearly deterministic, this evidence takes a universal form. For the two-point law, $m_\Delta(\theta) = 1 + \Delta\, \varphi(\theta) + O(\Delta^2)$ with $\varphi(\theta) = \int_0^1 \frac{(1+w)^\theta - 1}{w}\, dw - \theta$, so that $\chi^2(F_\Delta \| \mathrm{Unif}) \approx \Delta/2$ and $\mathrm{KL}(F_\Delta \| \mathrm{Unif}) \approx 0.209\, \Delta$ (Lemma B.3, Appendix B). The expansion reflects the fact that, for small Δ, the watermark is a small bump that at most doubles the density of pivots just below 1, coming from the rare events in which the model departs from its top choice. It has two consequences. First, the information per token is *linear* in Δ, which is the signature of a rare, strong signal, whereas a weak Gaussian mean shift δ has information of order δ^2. Second, a block of ℓ tokens has cumulant $\ell \Delta\, \varphi(\theta)$, which depends on the NTPs only through $\ell\Delta$, the expected number of *events* per segment. Since $\mathrm{KL} \approx 0.21\, \Delta$, "events per segment" is simply the small-freedom reading

of "evidence per segment." For moderate freedom, by contrast, every watermarked token carries some evidence (a top token chosen with probability 0.5 has pivot $V^{0.5}$, with mean 2/3), and the KL rather than the event count is the right measure.

5 Main results

We present the main results in three steps. We first show that tests which ignore token order cannot benefit from contiguity at all. We then describe a coarse phase diagram, which shows that segment-aware tests benefit once segments contain many events. Finally, we zoom in on the critical scale between these regimes, where we obtain a sharp boundary.

5.1 Order-blind tests cannot use contiguity

Our first result is negative. It says that a test which ignores token order performs no better under segment survival than under i.i.d. survival.

Proposition 5.1 (Order-blind barrier)**.** *Let $P_t \equiv P_n$ for all t, and suppose $n\epsilon_n^2\chi^2(F_{P_n}\|\mathrm{Unif}) \to 0$ (this reads $2p+q>1$ for P^{Δ_n} and $p>1/2$ for a fixed P, and $p>a/2$ in critical units in both cases). Every test invariant under permutations of $(Y_1,\dots,Y_n)$ has $R_n \to 1$, for every segment length $\ell_n \le \epsilon_n n$.*

The reason is that an order-blind test sees only the histogram of the pivots, which shuffling leaves unchanged. Such a test therefore performs on any arrangement of runs exactly as on a uniformly random set of the same size, which is the i.i.d. problem, and Li et al.'s lower bound applies. The proof, a second-moment calculation, is given in Appendix C. In particular, the sum test, token-level HC and Tr-GoF, which are the natural tests under i.i.d. survival, gain nothing from contiguity.

5.2 The coarse phase diagram

Segment-aware tests, in contrast, can exploit contiguity, and the next result shows exactly when they do on the polynomial scale.

Theorem 5.2 (Coarse phase diagram)**.** *Consider the coarse regime, with $\ell_n = n^\gamma$ and $p+\gamma<1$.*

(i) If $\ell_n\Delta_n = O(1)$ (in particular if $\gamma<q$) and $2p+q>1$, the problem is undetectable.

(ii) If $2p+q<1$, a global count of pivots above $1-\Delta_n$ has $R_n \to 0$.

(iii) If $\gamma>q$, a scan over block counts of pivots above $1-\Delta_n$, with blocks of length $\lfloor \ell_n/2 \rfloor$, has $R_n \to 0$.

Parts (ii) and (iii) hold uniformly over $\mathcal{P}_\Delta$. Hence, for $\gamma \neq q$, the detectable region is $\{2p+q<1\} \cup \{\gamma>q\}$ (Figure 1). On the region $\{2p+q>1,\ \gamma>q\}$, order-blind tests fail while the block scan succeeds.

Theorem 5.2 says that contiguity helps if and only if segments are longer than $1/\Delta$, that is, if they contain many events. Segments shorter than $1/\Delta$ rarely contain even one event, so there is nothing to aggregate and the i.i.d. boundary remains. Segments with polynomially many events, on the other hand, are each individually conspicuous, so a scan finds one as soon as any exist. Neither test needs to know (Δ,ℓ), since dyadic grids of thresholds and block lengths give the same conclusions (Appendix C). The boundary case $\gamma=q$ is the critical regime, to which we now turn.

5.3 The sharp boundary at the critical scale

The coarse diagram jumps from "contiguity is useless" to "any segment can be found." The transition occurs when one segment's expected log-likelihood ratio is of order $\log n$, which is enough to be suggestive but not individually conclusive. This mirrors the $\sqrt{2r\log p}$ signals from class, whose log-likelihood ratio $r\log p$ is also of order $\log p$.

To analyze this regime, we reduce the problem to a sparse mixture of blocks. Tile the text into $N = n^{a+o(1)}$ blocks aligned with the segments. Each block is then either all-null or all-watermarked, and its evidence is summarized by its log-likelihood ratio T. The problem becomes rare/weak detection over N "super-observations," a fraction ϵ of which carry signal, with a non-Gaussian law. On the $\log n$ scale, the block's law enters only through the limit $K(\theta) = \lim_n \log \mathbb{E}_0 e^{\theta T}/\log n$ of its null cumulant (Assumption A6 in Appendix D). Then $K(0) = K(1) = 0$. Let $t_0 = K'(0) < 0 < t_1 = K'(1)$, and let $I(t) = \sup_\theta\{\theta t - K(\theta)\}$ denote the large-deviation rate of $T/\log n$ under the null. We define

$$D(K, a) = \frac{a}{2} + \sup_{t \geq t_0} \Big[t - I(t) + \tfrac{1}{2}\min\{I(t), a\}\Big].$$

The test ψ^* computes a p-value π_k for each block's T_k and rejects if HC across the N block p-values (at dyadic levels) or a Bonferroni test on $\min_k \pi_k$ is large. Theorem D.1 in Appendix D shows that for a general sparse mixture of N blocks the problem is detectable by ψ^* if $p < D(K, a)$ and undetectable if $p > D(K, a)$.

The formula for D has a simple counting interpretation. Consider the number of blocks whose evidence exceeds tL. By large deviations the null count has mean $n^{a-I(t)}$, while the watermark adds about $n^{a-p-(I(t)-t)}$ such blocks. The count succeeds if the excess beats both 1 and the null standard deviation $n^{(a-I(t))/2}$, and optimizing over t gives $p < D$. HC searches over all thresholds at once, and the Bonferroni component covers thresholds where the null count is below 1. A Hellinger argument gives the matching lower bound. The condition $p < D$ is Cai and Wu's (2014) boundary in different units, and for Gaussian blocks it reduces to the Donoho–Jin boundary from class (Remark D.2).

To apply this result to watermarks, recall from Section 4 that a watermarked block's cumulant is the sum of its tokens' cumulants. We assume that this sum is the same for every block.

Assumption A5 (Block-homogeneous NTPs)**.** Segments are aligned to a grid of $N = n^{a+o(1)}$ blocks of length ℓ_n. Every block has the same multiset of NTPs, known to the detector (an oracle), and the block cumulant $\Lambda_n(\theta) = \sum_{t\in\text{block}} \log m_{P_t}(\theta)$ satisfies Assumption A6 with limit K.

Under this assumption the blocks are interchangeable, so the general theorem applies. There are two main instances. With *fixed NTPs*, $P_t \equiv P$ and $\ell_n = c\log n$, which gives $a = 1$ and $K = c\log m_P$. With *small freedom*, $P_t \equiv P^{\Delta_n}$ and $\ell_n\Delta_n = \kappa\log n$, which gives $a = 1 - q$ and $K = \kappa\varphi$ (Lemma B.3). The assumption ignores NTPs that vary across blocks, a case we probe numerically in Section 6.

Theorem 5.3 (Sharp boundaries for segment-surviving watermarks)**.** *Suppose Assumption A5 holds with limiting block cumulant K. Then the following hold.*

(a) ***Optimal boundary.*** *If $p < D(K, a)$, then ψ^* applied to the block log-likelihood ratios $T_k = \sum_{t\in block\ k} \log f_{P_t}(Y_t)$ has $R_n \to 0$. If $p > D(K, a)$, the problem is undetectable.*

(b) ***Saturation.*** *$D(K, a) \geq a$, i.e., every configuration with $p < a$ is detectable, if and only if $K'(1) \geq a$. Here $K'(1)L = (1 + o(1))\sum_{t\in block} \mathrm{KL}(F_{P_t}\|\mathrm{Unif})$ is one segment's expected log-likelihood ratio and $aL = (1 + o(1))\log N$.*

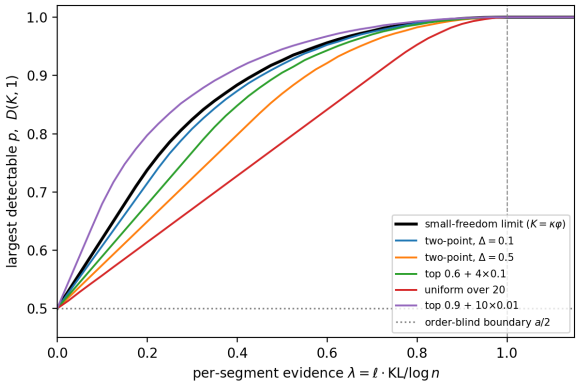


Figure 2: Sharp boundary $D(K, 1)$ for fixed NTPs against the per-segment evidence $\lambda = \ell \cdot \mathrm{KL}(F_P \| \mathrm{Unif})/\log n$. The ends are universal, with the order-blind boundary $1/2$ at $\lambda = 0$ and saturation exactly at $\lambda = a = 1$. The black curve is the small-freedom limit $K = \kappa\varphi$.

(c) ***Strict gain.*** *$D(K, a) > a/2$. If moreover $P_t \equiv P_n$ for all tokens, order-blind tests fail for every $p > a/2$ (Proposition 5.1).*

Theorem 5.3 identifies the boundary and shows that its two ends do not depend on the NTPs. Part (b) says that full detectability is a Bonferroni criterion. A single segment must stand out against the N places it could be, so its expected evidence must exceed $\log N$, and this holds for every NTP. Part (c) says that any positive evidence per segment moves the boundary past the line that order-blind tests cannot cross. Between these two ends, however, the shape of K matters (Figure 2). At equal per-segment KL, the boundaries of different NTPs differ by up to about 0.2, and evidence concentrated in rare, near-certain tokens (a peaked top choice with a tail of unlikely alternatives) is easier to detect than evidence spread evenly (a flat NTP). As in class, a statistic that looks for a few extreme blocks gains when per-block evidence is heavy-tailed.

In the small-freedom instance the boundary depends only on κ, the expected number of events per segment divided by $\log n$. It is then natural to ask how much is lost by a test that does not know the NTPs, and the following corollary answers this question.

Corollary 5.4 (Events per segment)**.** *Let $P_t \equiv P^{\Delta_n}$, $\Delta_n = n^{-q}$, $\ell_n \Delta_n = \kappa \log n$ and $a = 1 - q$, so that $K = \kappa\varphi$, and write $D_{\mathrm{full}}(\kappa, a) = D(\kappa\varphi, a)$. Then the following hold.*

(i) D_{full} is nondecreasing and lower semicontinuous in κ.

(ii) $D_{\mathrm{full}} \geq a$ if and only if $\kappa\varphi'(1) \geq a$.

(iii) (Key-only test.) For fixed $u > 0$, among tests based on the block counts $C_k(u) = \#\{t \in \text{block } k : Y_t > 1 - u\Delta_n\}$, the boundary is $D(K_u, a)$, with K_u the counts' cumulant (Lemma B.2), attained by ψ^* on these counts. Let $D_{\mathrm{cnt}}(\kappa, a) = \sup_u D(K_u, a) \leq D_{\mathrm{full}}(\kappa, a)$.

Part (ii) is Theorem 5.3(b) with $K'(1) = \kappa\varphi'(1)$, and for small κ a second-moment (χ^2) regime gives $D_{\mathrm{full}} = a/2 + \kappa/4$ exactly (Proposition E.1). The key-only test of part (iii) sees only approximately Poisson counts, so it pays for discarding where in the block each large pivot fell. On the other hand, it never uses the NTPs, and scanning thresholds and block lengths removes the need to know Δ_n and ℓ_n (Remark E.3). The price is small. At $a = 0.7$ and $\kappa = 1$ its boundary is 0.546, against 0.576 for the oracle, while every order-blind test is stuck at 0.35 (Table 2). Finally, the alignment in Assumption A5 is only a convenience. Running ψ^* on a few shifted grids per

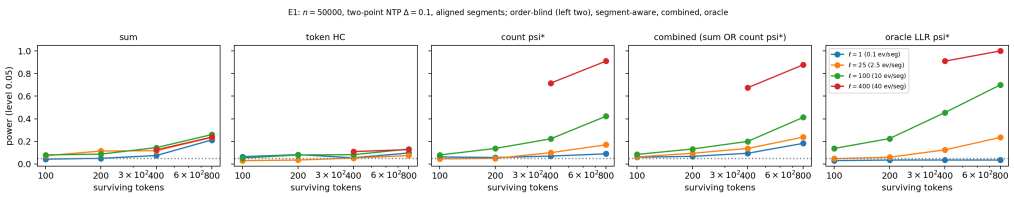


Figure 3: E1: power vs. number of surviving tokens, for four segment lengths.

block length, over a geometric grid of lengths, attains every $p < D(K, a)$ for arbitrary segments of unknown length (Proposition E.2, Appendix E).

6 Experiments

We now check these predictions in simulation. We simulate pivots under Assumption A1 and calibrate critical values by Monte Carlo at level 0.05 (standard error ≤ 0.03). We compare *order-blind* tests, namely the sum test $\sum_t S_t$ and token-level HC on $1 - Y_t$, with *segment-aware* tests, namely ψ^* on key-only block counts above $1 - 1.3\Delta$ or on oracle block log-likelihood ratios, and with a *combined* test that rejects if the sum test or the count-based ψ^* does, at Bonferroni-split levels. In E1, E2, E4 and E5 the block tests use the true segment length and a grid aligned with the segments, which is a best case for them. E3 removes both advantages, and the order-blind tests use neither.

Our first experiment (E1) isolates the effect of segment length. With $n = 5 \times 10^4$, $P_t \equiv P^{0.1}$ and segment lengths $\ell \in \{1, 25, 100, 400\}$, which correspond to 0.1 to 40 expected events per segment, we vary the number of surviving tokens (Figure 3). As Proposition 5.1 predicts, order-blind tests are flat in ℓ, with power at most 0.26. Segment-aware tests, in contrast, improve sharply with the number of events per segment. With 800 survivors, the oracle's power rises from 0.04 to 1.00 and the key-only test's from 0.09 to 0.91 as ℓ goes from 1 to 400.

The second experiment (E2) traces a finite-n version of the phase diagram, with $n = 2^{18}$, $\ell = N = 512$ ($a = 1/2$), $\Delta = \kappa \log n/\ell$ and round$(n^{-p}N)$ segments (Figure 4 in Appendix F). Token HC detects only near the order-blind boundary $p = a/2$, whereas segment-aware power extends far beyond it and grows with κ. At $\kappa \geq 2$, the oracle keeps power 0.69–0.96 even at $p = 0.45$ (two signal segments), where token HC has power at most 0.10. The 50%-power contour lies below the asymptotic curve, as expected at $\log n \approx 12.5$, but it has the predicted shape.

The remaining experiments, reported in Appendix F, move outside the sandbox. When segments are placed at random positions (E3), a single grid drops from power 0.84 to 0.53, while shifted grids (0.68) or a length-adaptive scan (0.71) recover most of the loss. With heterogeneous NTPs and unknown Δ (E4), the key-only test rises from power 0.08 to 1.00 as ℓ goes from 1 to 400, while order-blind tests stay between 0.15 and 0.40. Finally, E5 checks essay-length text ($n = 2000$). With sentence blocks ($N = 80$), saturation needs $\log N \approx 4.4$ nats per segment. At $\Delta = 0.4$ a sentence carries about this much, and four surviving sentences give the block tests power 0.89 (oracle) and 0.62 (key-only), against 0.29 for the sum test, whereas at $\Delta = 0.1$ the gain appears only for paragraphs. Since the sum test is best under i.i.d. survival, the combined test is a good default in practice.

7 Discussion

Our results suggest a simple practical reading. The many-segment regime of the sharp theory fits sentence-level partial retention, where a writer keeps a modest fraction of AI sentences verbatim and so leaves many short runs. Pasting two or three AI paragraphs into an essay corresponds instead to the endpoint $p \to a$, where the answer is the saturation criterion, so that a passage is detectable once its expected log-likelihood ratio exceeds $\log N$. Low-entropy content such as code or factual answers needs much longer runs (Theorem 5.2(i)).

These conclusions hold within the sandbox of Section 3 (Table 1). In particular, we adopt Li et al.'s pivot model, posit runs rather than model the editing process, and assume the same known NTPs in every block for the sharp results. All experiments are synthetic, and E1–E4 use 5×10^4 to 2.6×10^5 tokens, far longer than an essay (E5 checks $n = 2000$), with NTPs that are illustrative rather than measured. The most immediate next step is therefore a real-data study with an open model (e.g., GPT-2) and our own key, which would place news text, the course's MADStat abstracts and code on the axes of Figure 2. Localizing segments and extending the theory to green/red-list watermarks are natural further directions.

A Related work, tools, and the sandbox at a glance

Related work. Detecting signals on unknown intervals is a classical problem. Scan statistics maximize a local statistic over windows, Arias-Castro, Donoho and Huo (2005) derived optimal thresholds for such geometric signals, and Cai, Jeng and Li (2012) studied many short segments in genomics. In watermarking, WinMax (Kirchenbauer et al., 2024) scans windows for pasted watermarked passages, while WaterSeeker (Pan et al., 2025) and adaptive segmentation tests (Li, Liu and Li, 2025) localize watermarked segments. Xie et al. (2025) design watermarks that localize post-generation edits, Kuditipudi et al.'s (2024) edit-distance detector aligns text with the key sequence to tolerate insertions and deletions, and Li et al. (2025c) estimate the watermarked proportion under the i.i.d. mixture. None of these gives detection boundaries for contiguously surviving Gumbel-max watermarks or compares with the i.i.d.-optimal order-blind tests. Our proofs use the standard tools behind the undetectable region. These are second-moment and Hellinger lower bounds (Ingster, 1997; Donoho and Jin, 2004), Cai and Wu's (2014) boundary for general sparse mixtures, and large deviations on the $\log n$ scale (Dembo and Zeitouni, 1998), which we review next.

Lower bounds: second moments and Hellinger distance. The impossibility (undetectable) side of a phase diagram comes from Ingster (1997) and Donoho and Jin (2004). The idea is to place a random prior on where the signals are and show that the resulting mixture is statistically indistinguishable from the null. There are two standard ways to show this. The *second-moment method* shows that if the likelihood ratio Λ has $\mathbb{E}_0\Lambda^2 \to 1$, then the total variation distance tends to 0 and no test works. A *Hellinger-distance* bound uses the fact that Hellinger distance adds over independent coordinates, so it suffices to bound the distance contributed by one coordinate. The second moment is cleanest when the signal is not too strong, whereas the Hellinger route, combined with truncation, handles strong but rare signals.

General sparse mixtures. Donoho–Jin's boundary is for Gaussian signals. Cai and Wu (2014) characterize the boundary for an arbitrary null Q and alternative G in the mixture $(1-\epsilon)Q + \epsilon G$. The boundary is determined by a single function, namely the log-likelihood ratio $\log \frac{dG}{dQ}$ evaluated

at the null quantile with tail probability N^{-s}, viewed as a function of s. They also show that HC attains it adaptively. We need this generalization because our "observations" will be blocks of tokens, which are far from Gaussian.

Large deviations on the log scale. In class we wrote $\bar{\Phi}(\sqrt{2q\log p}) = L_p p^{-q}$, since only the polynomial order of a tail probability matters. Large-deviations theory generalizes this to sums of many small terms. If T is a sum whose log moment-generating function satisfies $\log \mathbb{E}e^{\theta T} \approx K(\theta)\log n$, then the Cramér and Gärtner–Ellis theorems (Dembo and Zeitouni, Ch. 2) give

$$\mathbb{P}(T > t\log n) = n^{-I(t)+o(1)}, \qquad I(t) = \sup_{\theta}\{\theta t - K(\theta)\}.$$

The rate I is the Legendre transform of the cumulant K. Intuitively, the cheapest way for T to be large is to "tilt" each summand, and I is the cost of the best tilt. We use this to compute the tail of a block's evidence on the $\log n$ scale.

In the sandbox	Outside the sandbox
independent pivots, fixed NTPs (A1)	NTPs depending on edited context; repeated keys
runs of length $\geq \ell$ covering $\geq \epsilon n$ tokens (A2)	the editing process; partial survival in a run; localization
Δ-regular NTPs (A3); same NTPs in every block for sharp results (A5)	NTPs varying across blocks (probed by E4)
power-law calibrations, $n \to \infty$ (A4)	finite-n constants; calibration at tiny false-positive rates
pivot-only tests	tests that also use the text's likelihood

Table 1: The theoretical sandbox at a glance.

B Pivot lemmas and proofs for Sections 2 and 4

The count tests use two further facts about pivots, namely how much mass a watermarked token puts just below 1 and the law of a block's count of large pivots.

Lemma B.1 (Excess tail mass)**.** *(a) If $P \in \mathcal{P}_\Delta$, $\Delta \leq 1/2$ and $0 < s \leq \Delta$, then $\mathbb{P}_P(Y > 1-s) \geq 2(1-e^{-1})s > 1.26\,s$, while $\mathbb{P}_{\mathrm{Unif}}(Y > 1-s) = s$.*

(b) For P^Δ and fixed $u > 0$: $r_u(\Delta) \equiv \mathbb{P}_{P^\Delta}(Y > 1-u\Delta) = \Delta\rho_u + O(\Delta^2)$, where $\rho_u = u + 1 - e^{-u}$.

In words, in a window of width $s \leq \Delta$ just below 1, a watermarked token puts at least 1.26 times the uniform mass, however its freedom Δ is spread over alternatives. For the two-point law, the excess $\Delta(1-e^{-u})$ saturates at Δ once the window is a few multiples of Δ wide. A wider window only adds null noise, which is why the best counting threshold sits near $1 - 1.3\Delta$.

Lemma B.2 (Block counts)**.** *For $u > 0$ and $s = u\Delta$, let $C = \#\{t \leq \ell : Y_t > 1-s\}$. Then $C \sim \mathrm{Bin}(\ell, s)$ under H_0, and $C \sim \mathrm{Bin}(\ell, r_u(\Delta))$ for ℓ tokens with $P_t \equiv P^\Delta$. If $\ell\Delta = \kappa L$, the count's log-likelihood ratio T_u satisfies $\log \mathbb{E}_0 e^{\theta T_u} = L\big(K_u(\theta) + o(1)\big)$, where*

$$K_u(\theta) = \kappa\Big[u\big((\rho_u/u)^\theta - 1\big) - \theta(\rho_u - u)\Big].$$

In words, a block's count of large pivots is approximately Poisson, with mean $\kappa u L$ under the null and $\kappa \rho_u L$ under the watermark. K_u is the corresponding Poisson cumulant on the $\log n$ scale.

Lemma B.3 (Per-token evidence for small freedom)**.** *Let $m_\Delta = m_{P^\Delta}$ and $\varphi(\theta) = \int_0^1 \frac{(1+w)^\theta - 1}{w}\, dw - \theta$. Uniformly over compact sets of $\theta \in \mathbb{R}$,*

$$m_\Delta(\theta) = 1 + \Delta\, \varphi(\theta) + O(\Delta^2).$$

Hence $\chi^2(F_\Delta \| \mathrm{Unif}) = \Delta/2 + O(\Delta^2)$ and $\mathrm{KL}(F_\Delta \| \mathrm{Unif}) = \Delta\varphi'(1) + o(\Delta)$. The function φ is smooth and strictly convex, with $\varphi(0) = \varphi(1) = 0$, $\varphi(2) = \frac{1}{2}$, $\varphi'(1) = \frac{\pi^2}{12} + 2\log 2 - 2 \approx 0.209$ and $\varphi'(2) = \frac{\pi^2}{12} + 4\log 2 - \frac{11}{4} \approx 0.845$.

Proof of Lemma 2.1. Let $E_w = -\log U_{t,w} \sim \mathrm{Exp}(1)$, so $w_t = \mathrm{argmin}_w E_w / P_{t,w}$. The minimum M of independent exponentials with rates $P_{t,w}$ is $\mathrm{Exp}(1)$ and is independent of the argmin, which equals w with probability $P_{t,w}$. Hence $Y_t = e^{-P_{t,w_t} M} = V^{P_{t,w_t}}$ with $V = e^{-M} \sim \mathrm{Unif}(0,1)$. For a token chosen without the key, the key is independent of the token, so $Y_t \sim \mathrm{Unif}(0,1)$. □

Proof of Lemma B.1. (a) $1 - F_P(1-s) = \sum_w P_w\big(1 - (1-s)^{1/P_w}\big) \geq \sum_w P_w(1 - e^{-s/P_w}) \geq (1 - e^{-1}) \sum_w \min(s, P_w)$, using $1 - e^{-x} \geq (1 - e^{-1}) \min(x, 1)$. It remains to show $\sum_w \min(s, P_w) \geq 2s$. If some $P_{w_0} \geq s$, that token contributes s, and the others have total mass $1 - P_{w_0} \geq \Delta \geq s$, so by subadditivity of $x \mapsto \min(s, x)$ they contribute $\geq s$. Otherwise every term equals P_w, and the sum is $1 \geq 2s$. (b) $r_u(\Delta) = 1 - (1-\Delta)(1 - u\Delta)^{1/(1-\Delta)} - \Delta(1 - u\Delta)^{1/\Delta}$, with $(1 - u\Delta)^{1/(1-\Delta)} = 1 - u\Delta + O(\Delta^2)$ and $(1 - u\Delta)^{1/\Delta} = e^{-u} + O(\Delta)$. □

Proof of Lemma B.3. Substituting $y = e^{-x}$ and factoring $e^{-\alpha x}$,

$$m_\Delta(\theta) - 1 = \underbrace{\int_0^\infty e^{-x}(e^{-\alpha\theta x} - 1)\, dx}_{= -\alpha\theta/(1+\alpha\theta)} + \int_0^\infty e^{-(1+\alpha\theta)x} \big[(1 + e^{-(\beta-\alpha)x})^\theta - 1\big]\, dx.$$

The first term is $-\theta\Delta + O(\Delta^2)$. In the second, set $v = (\beta - \alpha)x$, so that $1/(\beta - \alpha) = \Delta(1-\Delta)/(1 - 2\Delta) = \Delta + O(\Delta^2)$ and $c \equiv (1 + \alpha\theta)/(\beta - \alpha) = O(\Delta)$. Since $|e^{-cv} - 1| \leq cv$ and $|(1 + e^{-v})^\theta - 1| \leq |\theta| \max(1, 2^{\theta-1}) e^{-v}$, the second term equals $\Delta \int_0^\infty [(1 + e^{-v})^\theta - 1]\, dv + O(\Delta^2)$. Substituting $w = e^{-v}$ identifies the integral with $\varphi(\theta) + \theta$. The constants are uniform over compact sets (for negative θ we need $1 + \alpha\theta > 0$, true for small Δ). Then $\chi^2 = m_\Delta(2) - 1$, and $\mathrm{KL}(F_\Delta \| \mathrm{Unif}) = m'_\Delta(1)$. Uniform convergence of the convex functions $(m_\Delta - 1)/\Delta$ to the differentiable φ implies convergence of their derivatives, which gives the KL statement. The derivative values are elementary integrals, and $\varphi''(\theta) = \int_0^1 (1+w)^\theta \log^2(1+w)\, w^{-1} dw > 0$. □

Proof of Lemma B.2. $T_u = \lambda C + \ell \log \frac{1-r}{1-s}$ with $\lambda = \log \frac{r(1-s)}{s(1-r)} = \log(\rho_u/u) + O(\Delta)$. Then $\log \mathbb{E}_0 e^{\theta T_u} = \ell \log(1 - s + s e^{\theta\lambda}) + \theta\ell \log \frac{1-r}{1-s} = \ell s(e^{\theta\lambda} - 1) - \theta\ell(r - s) + O(\ell\Delta^2)$, using Lemma B.1(b). Substitute $\ell s = u\kappa L$ and $\ell(r - s) = (\rho_u - u)\kappa L + O(L\Delta)$. K_u is smooth and strictly convex, with $K_u(0) = K_u(1) = 0$. □

C Proofs of Proposition 5.1 and Theorem 5.2

Proof of Proposition 5.1. Fix $S \in \mathcal{S}_n(\epsilon, \ell)$ with $|S| = k = \ell\lceil \epsilon n/\ell \rceil = (1 + o(1))\epsilon n$, a union of runs of length ℓ. By invariance and Assumption A1, $\mathbb{P}_S(\psi = 0) = \mathbb{P}_{\Pi S}(\psi = 0)$ for every permutation Π, so it equals the type II error under the prior π_k that picks a uniformly random k-subset. Hence

$R_n(\psi) \geq 1 - \mathrm{TV}(\mathbb{P}_0, \mathbb{P}_{\pi_k})$. Write $\chi^2 = \chi^2(F_{P_n} \| \mathrm{Unif})$. The likelihood ratio $\Lambda = \mathbb{E}_{\pi_k} \prod_{t \in S} f_{P_n}(Y_t)$ satisfies $\mathbb{E}_0 \Lambda^2 = \mathbb{E}(1+\chi^2)^H$, where $H = |S_1 \cap S_2|$ for independent $S_1, S_2 \sim \pi_k$ is hypergeometric. By Hoeffding (1963, Thm. 4), $\mathbb{E}g(H) \leq \mathbb{E}g(\mathrm{Bin}(k, k/n))$ for convex g. So $\mathbb{E}_0\Lambda^2 \leq (1 + k\chi^2/n)^k \leq \exp(k^2\chi^2/n) = \exp\big((1+o(1))n\epsilon^2\chi^2\big) \to 1$, and $\mathrm{TV} \leq \frac{1}{2}(\mathbb{E}_0\Lambda^2 - 1)^{1/2} \to 0$. □

Proof of Theorem 5.2. (i) Let π mark each block of the aligned grid of length ℓ_n independently with probability $\epsilon' = 2\epsilon_n$. Then $|S| = \ell_n \mathrm{Bin}(N, \epsilon')$, and by Chernoff $\pi(\mathcal{S}_n(\epsilon, \ell)) \to 1$ because $N\epsilon_n = n^{1-p-\gamma+o(1)} \to \infty$. For any ψ,

$$R_n(\psi) \geq \mathbb{P}_0(\psi = 1) + \mathbb{P}_\pi(\psi = 0) - \pi(\mathcal{S}_n^c) \geq 1 - \mathrm{TV}(\mathbb{P}_0, \mathbb{P}_\pi) - o(1).$$

The likelihood ratio is $\prod_k (1 - \epsilon' + \epsilon' W_k)$, with independent block ratios satisfying $\mathbb{E}_0 W_k^2 = (1+\chi^2)^\ell$. Hence

$$\mathbb{E}_0\Big[\prod_k (1 - \epsilon' + \epsilon' W_k)^2\Big] = \big(1 + \epsilon'^2[(1+\chi^2)^\ell - 1]\big)^N \leq \exp\big(4N\epsilon^2(e^{\ell\chi^2} - 1)\big) \leq \exp\big(Cn\epsilon^2\Delta\big) \to 1,$$

using $\ell\chi^2 = O(1)$ and $e^x - 1 \leq C'x$ on bounded sets. (ii) The global count test rejects when $\#\{t : Y_t > 1 - \Delta_n\} > n\Delta_n + (n\Delta_n)^{1/2}L$. Under H_0 the count is $\mathrm{Bin}(n, \Delta)$, so by Chebyshev the type I error is $\leq L^{-2}$. Under H_1 it is a sum of independent indicators which, by Lemma B.1(a), stochastically dominates $B = \mathrm{Bin}(n - |S|, \Delta) + \mathrm{Bin}(|S|, 1.26\Delta)$. Here $\mathbb{E}B - n\Delta = 0.26|S|\Delta$ and $\mathrm{Var}\, B \leq 2n\Delta$. Since $|S|\Delta/(\sqrt{n\Delta}L) \geq n^{(1-2p-q)/2}/L \to \infty$, Chebyshev applied to B gives type II error $\to 0$. (iii) The block scan tiles $[n]$ by blocks of length $b = \lfloor \ell_n/2 \rfloor$, counts $C_k = \#\{t \in \text{block } k : Y_t > 1 - \Delta_n\}$, and rejects if $\max_k C_k > \lambda + \sqrt{3\lambda L}$, where $\lambda = b\Delta_n$. Every interval of length $\geq \ell_n$ contains a full block. Under H_0, $C_k \sim \mathrm{Bin}(b, \Delta)$. Bernstein's inequality with $x = \sqrt{3\lambda L} = o(\lambda)$ gives $\mathbb{P}(C_k > \lambda + x) \leq \exp\big(-\frac{x^2}{2(\lambda + x/3)}\big) = n^{-3/2+o(1)}$, and a union bound over $\leq n$ blocks controls the type I error. A full signal block has $C_k \succeq \mathrm{Bin}(b, 1.26\Delta)$. Its mean excess 0.26λ dominates both x and its standard deviation, since $\lambda \asymp n^{\gamma-q} \gg L$. □

Adaptivity. Without knowledge of (Δ_n, ℓ_n), apply (ii) over thresholds $s \in \{2^{-j}\}$ and (iii) over $(s, b) \in \{2^{-j}\} \times \{2^j\}$, with Bonferroni corrections. Some grid point has $s \in [\Delta/2, \Delta]$ and $b \in [\ell/4, \ell/2]$, where the same arguments go through with constants halved. The type I errors sum to $O(\log n/L^2) + O(\log^2 n \cdot n^{-1/2+o(1)}) \to 0$.

D Sharp boundary for sparse block mixtures

There are N independent blocks with law Q_n under the null and G_n if watermarked. The null hypothesis H_0 is that no block is watermarked, and the alternative H_1 is that at least ϵN are. Let $T = \log \frac{dG_n}{dQ_n}$ and $\Lambda_n(\theta) = \log \mathbb{E}_{Q_n} e^{\theta T}$.

Assumption A6 (Block cumulants)**.** $N = n^{a+o(1)}$ with $a \in (0, 1]$, $\epsilon = n^{-p+o(1)}$ with $0 < p < a$, and $\Lambda_n(\theta)/L \to K(\theta)$ for every θ in an interval (θ_-, ∞) with $\theta_- \in [-\infty, 0)$. On this interval K is finite, differentiable and strictly convex, and, if $\theta_- > -\infty$, $K(\theta) \to \infty$ as $\theta \downarrow \theta_-$. Let $t_+ = \lim_{\theta\to\infty} K'(\theta) \in (K'(1), \infty]$ (finite, e.g., when per-token log-likelihood ratios are bounded, as for a fixed NTP).

Then $K(0) = K(1) = 0$. Let $t_0 = K'(0) < 0 < t_1 = K'(1)$ and $I(t) = \sup_\theta\{\theta t - K(\theta)\}$. Then $I(t) - t$ is the corresponding rate under G_n, and $I = +\infty$ on (t_+, ∞). Define

$$D(K, a) = \frac{a}{2} + \sup_{t \geq t_0} \Big[t - I(t) + \tfrac{1}{2}\min\{I(t), a\}\Big].$$

The test ψ^* (HC across blocks). Let $\pi_k = Q_n(T > T_k) + V_k\, Q_n(T = T_k)$, with $V_k \overset{\text{iid}}{\sim} \text{Unif}(0,1)$, be randomized block p-values, exactly uniform under H_0 even for discrete T (unrandomized discrete p-values would bias HC's numerator). Let $\text{HC}_N^{\text{dy}} = \max_{1 \le j \le \log_2 N} \big(\#\{k : \pi_k \le 2^{-j}\} - N2^{-j}\big)/\sqrt{N2^{-j}}$ be HC at dyadic levels. The test ψ^* rejects if $\text{HC}_N^{\text{dy}} > L^2$ or $\min_k \pi_k < (NL)^{-1}$ (a Bonferroni component). It needs the null block law Q_n but not (p, ϵ).

Theorem D.1 (Sharp boundary for sparse block mixtures)**.** *Suppose Assumption A6 holds. If $p > D(K,a)$ the problem is undetectable, and if $p < D(K,a)$ then $R_n(\psi^*) \to 0$.*

Remark D.2 (Relation to known results)**.** After the change of units $s = I(t)/a$, $\varsigma(s) = t/a$, $\beta = p/a$, the condition $p < D(K,a)$ is exactly Cai and Wu's (2014) boundary $\beta < \frac{1}{2} + \sup_s\{\varsigma(s) - s + \frac{1}{2}(s \wedge 1)\}$. Their assumptions are stated in terms of log-likelihood ratios at null quantiles, whereas ours, convergence of block cumulants, are immediate for sums over tokens. For Gaussian blocks, $K(\theta) = r\theta(\theta - 1)$, and with $a = 1$ the condition reduces to $r > \rho_{\text{detect}}(p)$, the Donoho–Jin boundary from class (checked numerically). So Theorem D.1 contains the class result as a special case.

Throughout, $I_1(t) \equiv \sup_\theta\{\theta t - K(\theta+1)\} = I(t) - t$ is the rate function under G_n, since $\log \mathbb{E}_{G_n} e^{\theta T} = \Lambda_n(\theta + 1)$. Define

$$g(t) = a - I_1(t) - \max\Big\{0, \tfrac{a - I(t)}{2}\Big\},$$

so that $D(K,a) = \sup_{t \ge t_0} g(t)$. In the counting heuristic, $g(t) > p$ says that the excess count at threshold tL beats both 1 and the null standard deviation.

(a) Lower bound. Let π mark blocks independently with probability $\epsilon' = 2\epsilon$. Then $\pi(|J| \ge \epsilon N) \to 1$, and as above $\inf_\psi R_n \ge 1 - \text{TV}(\mathbb{P}_0, \mathbb{P}_\pi) - o(1)$. With $H^2(P,Q) = \int(\sqrt{dP} - \sqrt{dQ})^2$, we have $\text{TV} \le H$ and $H^2(\otimes P_k, \otimes Q_k) \le \sum_k H^2(P_k, Q_k)$. Here $\mathbb{P}_\pi = M^{\otimes N}$ with $M = (1-\epsilon')Q_n + \epsilon' G_n$, and $H^2(Q_n, M) = \mathbb{E}_Q[(\sqrt{1+x} - 1)^2]$ with $x = \epsilon'(e^T - 1) \ge -1$. We use $(\sqrt{1+x} - 1)^2 \le \min(x^2, |x|)$. On $\{T \le y\}$, with $y = \log(1/\epsilon')$, use $x^2 \le \epsilon'^2(e^{2T} + 1)$. On $\{T > y\}$, use $|x| \le \epsilon' e^T$ together with $\mathbb{E}_Q[e^T 1\{T > y\}] = G_n(T > y)$. Hence

$$NH^2(Q_n, M) \le N\epsilon'^2 + N\epsilon'^2\, \mathbb{E}_Q[e^{2T} 1\{T \le y\}] + N\epsilon'\, G_n(T > y).$$

For $\theta \in [0,2]$, $e^{2T} 1\{T \le y\} \le e^{\theta T + (2-\theta)y}$, and for $\theta \ge 0$, $1\{T > y\} \le e^{\theta(T-y)}$. With $y = pL - \log 2$, and Assumption A6 applied at a θ nearly attaining each infimum (if $p > t_+$ then $E_3 = -\infty$), the three terms are at most $n^{E_j + o(1)}$, where

$$E_1 = a - 2p, \qquad E_2 = a - 2p + \inf_{\theta \in [0,2]}\{K(\theta) + (2-\theta)p\}, \qquad E_3 = a - p - I_1^+(p),$$

with $I_1^+(p) = \sup_{\theta \ge 0}\{\theta p - K(\theta + 1)\}$. If $p > D$, Lemma D.3 gives $\max_j E_j < 0$, so $\text{TV} \to 0$.

Lemma D.3 (Duality)**.** *If $\max\{E_1, E_2, E_3\} \ge 0$, then $p \le D(K,a)$.*

Proof. We exhibit $t \ge t_0$ with $g(t) \ge p$.

- $E_1 \ge 0$: take $t = t_1$, where $I_1 = 0$ and $I = t_1$; then $g(t_1) = \min\{a, (a + t_1)/2\} > a/2 \ge p$.
- $E_2 \ge 0$: let $\tilde{I}(t) = \sup_{\theta \in [0,2]}\{\theta t - K(\theta)\}$, so that $2t - \tilde{I}(t) = \inf_{\theta \in [0,2]}\{K(\theta) + (2-\theta)t\}$. For each $\theta \in [0,2]$ the bracket is nondecreasing in t, hence so is the infimum, and $\inf_{\theta \in [0,2]}\{K(\theta) + (2-\theta)p\} = \sup_{t \le p}\{2t - \tilde{I}(t)\}$. The map $t \mapsto 2t - \tilde{I}(t)$ is nondecreasing (slope $2 - \theta(t) \ge 0$), so the supremum is attained at $t^\star = \min\{p, K'(2)\} \in (0, K'(2)]$, where $\tilde{I}(t^\star) = I(t^\star)$. Thus $a - 2p + 2t^\star - I(t^\star) \ge 0$. Then $a - p - I_1(t^\star) = a - p - I(t^\star) + t^\star \ge p - t^\star \ge 0$, and $a - p - I_1(t^\star) - \frac{a - I(t^\star)}{2} = \frac{1}{2}(a - I(t^\star) + 2t^\star - 2p) \ge 0$; so $g(t^\star) \ge p$.

- $E_3 \geq 0$ and $p \geq t_1$: then $I_1^+(p) = I_1(p)$ and $I(p) = I_1(p) + p$. With $x = a - p - I_1(p) \geq 0$, we have $\frac{a-I(p)}{2} = \frac{x}{2} \leq x$, so $g(p) \geq p$.
- $E_3 \geq 0$ and $p < t_1$: then $g(t_1) = \min\{a, (a+t_1)/2\} > p$, because $p < \min\{a, t_1\}$.

□

Lemma D.4. *g is continuous on (t_0, t_+), equals $-\infty$ on (t_+, ∞), and is strictly increasing on $[t_0, 0]$. Hence $D(K, a) = \sup_{t>0} g(t)$, and if $p < D$ there is $t > 0$ with $g(t) > p$ and $I(t) \neq a$.*

Proof. I is convex and finite on (t_0, t_+), hence continuous there, and so is g. On (t_+, ∞) we have $I = +\infty$ and $g = -\infty$. Write $g(t) = a - I(t) + t - \max\{0, (a - I(t))/2\}$, and note $I'(t) = \theta(t)$ with $K'(\theta(t)) = t$. The slope of g is $1 - \theta(t)$ where $I(t) > a$ and $1 - \theta(t)/2$ where $I(t) < a$. On $[t_0, 0]$ we have $0 \leq \theta(t) \leq \theta(0) < 1$, because $K'(1) = t_1 > 0$. So both slopes are positive and g is strictly increasing there. Since I is strictly increasing on (t_0, t_+), a t with $g(t) > p$ can be taken in $(0, t_+)$, where $I(t) < \infty$ (if $t = t_+ < \infty$, move t slightly left, using continuity of the closed convex I on its domain), and the single point with $I(t) = a$ can be avoided by continuity. □

(b) Upper bound. By Assumption A6, T/L satisfies the Gärtner–Ellis hypotheses under Q_n with speed L: the limit K is finite and differentiable on the open interval $(\theta_-, \infty) \ni 0$ and, if $\theta_- > -\infty$, blows up at θ_- (so $K' \to -\infty$ there, by convexity); hence it is essentially smooth (Dembo and Zeitouni, Thm. 2.3.6). Gärtner–Ellis also needs the limit for every θ: for $\theta \leq \theta_- < \theta' < \theta''$, convexity gives $\Lambda_n(\theta) \geq \Lambda_n(\theta') + (\theta' - \theta)\frac{\Lambda_n(\theta') - \Lambda_n(\theta'')}{\theta'' - \theta'}$, and dividing by L and letting $\theta' \downarrow \theta_-$ shows $\Lambda_n(\theta)/L \to +\infty$. So the limit is K extended by $+\infty$, which is lower semicontinuous because $K \to \infty$ at θ_-. This gives a large deviation principle with rate I; under G_n the limit is $K(\cdot + 1)$, with rate I_1. Hence $h_0(t) \equiv Q_n(T \geq tL) = n^{-I(t)+o(1)}$ for $t > t_0$. Likewise $h_1(t) \equiv G_n(T \geq tL) = n^{-I_1(t)+o(1)}$ for $t > t_1$ and $h_1(t) \to 1$ for $t < t_1$; in all cases $h_1 \geq n^{-I_1(t)+o(1)}$. By Lemma D.4, fix $t > 0$ with $g(t) > p$ and $I(t) \neq a$. Note that $\{T_k \geq tL\} \subseteq \{\pi_k \leq h_0\}$, since $\pi_k \leq Q_n(T \geq T_k) \leq Q_n(T \geq tL)$; and under H_0 the π_k are i.i.d. $\mathrm{Unif}(0,1)$, so null counts below any level α are exactly $\mathrm{Bin}(\cdot, \alpha)$.

Case $I(t) < a$. Then $h_0 \gg 1/N$; choose a dyadic level $\alpha \in [h_0, 2h_0]$. The count $X_\alpha = \#\{k : \pi_k \leq \alpha\}$ dominates $\mathrm{Bin}(N - |J|, \alpha) + \mathrm{Bin}(|J|, h_1)$, so $X_\alpha - N\alpha \geq |J|(h_1 - \alpha) - O_P(\sqrt{N\alpha} + \sqrt{|J|h_1})$. Since $h_1/h_0 = n^{t+o(1)} \to \infty$ and $|J| \geq \epsilon N$,

$$\frac{|J|(h_1 - \alpha)}{\sqrt{N\alpha}} \geq n^{a-p-I_1(t)-(a-I(t))/2+o(1)} = n^{g(t)-p+o(1)} \gg L^2,$$

and $|J|h_1 \to \infty$ because $g(t) > p$ forces $a - p - I_1(t) > 0$.

Case $I(t) > a$. The number of signal blocks with $T_k \geq tL$ stochastically dominates $\mathrm{Bin}(|J|, h_1)$, whose mean is $\geq n^{g(t)-p+o(1)} \to \infty$. So with probability $\to 1$ some block has $\pi_k \leq h_0 = n^{-I(t)+o(1)} < (NL)^{-1}$.

Type I error. For each dyadic level $N\alpha \geq 1$, and Bernstein gives $\mathbb{P}_0\big(\mathrm{Bin}(N, \alpha) - N\alpha > x\sqrt{N\alpha}\big) \leq \exp\big(-\frac{x^2}{2(1+x/3)}\big) \leq e^{-x/2}$ for $x \geq 3/2$. With $x = L^2$, a union bound over $\log_2 N$ levels gives $o(1)$, and $\mathbb{P}_0(\min_k \pi_k < (NL)^{-1}) \leq 1/L$.

E Proofs of Theorem 5.3 and Corollary 5.4; further results

Proof of Theorem 5.3. (a) Block k has likelihood ratio e^{T_k} between its watermarked law $\bigotimes_{t \in \text{block } k} F_{P_t}$ and its null law $\mathrm{Unif}^{\otimes \ell}$, with $T_k = \sum_{t \in \text{block } k} \log f_{P_t}(Y_t)$. Hence $(T_k)_k$ is sufficient for the set J of watermarked blocks, so the total variation distance between the null and any mixture over J, and

so the lower bound, can be computed on (T_k), and ψ^* is a function of (T_k). The T_k are independent, and since every block has the same multiset of NTPs, their null and alternative laws do not depend on k (blocks with the NTPs in different orders have permuted product laws, but the same law of T_k). So Theorem D.1 applies to (T_k). By independence (Assumption A1), $\Lambda_n(\theta) = \sum_{t\in\text{block}} \log m_{P_t}(\theta)$, which satisfies Assumption A6 by Assumption A5. Theorem D.1 now applies, and the test ψ^* uses Q_n, which the oracle knows. (b) $g(t) \le a - I_1(t)$, and $I_1 \ge 0$ vanishes only at $t_1 = K'(1)$. So $\sup g \ge a$ forces $t = t_1$, where $g(t_1) = a - \max\{0, (a-t_1)/2\}$, and this is at least a iff $t_1 \ge a$. The supremum is attained because $g(t) \to -\infty$ as $t \to \infty$. For the interpretation, note that the functions Λ_n/L are convex and converge pointwise to K, which is differentiable at 1. Hence $\Lambda_n'(1)/L \to K'(1)$, where $\Lambda_n'(1) = \sum_{t\in\text{block}} \mathrm{KL}(F_{P_t}\|\mathrm{Unif})$, and $\log N = (a+o(1))L$. (c) K is strictly convex with $K(0) = K(1) = 0$, so $t_1 = K'(1) > 0$ and $g(t_1) = a - \max\{0, (a-t_1)/2\} > a/2$. The statement about order-blind tests is Proposition 5.1. For fixed P, χ^2 is a constant and $n\epsilon^2 = n^{1-2p}$, while for P^{Δ_n} we have $\chi^2 \sim \Delta_n/2$ (Lemma B.3) and $n\epsilon^2\Delta_n = n^{1-2p-q}$. □

Proof of Corollary 5.4. By Lemma B.3, $\Lambda_n(\theta) = \ell \log m_\Delta(\theta) = \ell\Delta\varphi(\theta) + O(\ell\Delta^2) = L\kappa\varphi(\theta) + o(L)$ for every $\theta \in \mathbb{R}$ ($m_\Delta(\theta)$ is finite once $\theta > -1/\alpha$, and $\alpha \to 0$). The limit $K = \kappa\varphi$ is finite, differentiable and strictly convex on $\mathbb{R}$ with $K' \to \infty$, so Assumption A5 holds with $a = 1 - q$. (i) For monotonicity, discard the last $(1 - \kappa/\kappa')\ell$ tokens of each block. This reduces a κ'-problem to a κ-problem with the same (a, p), since $\log n$ changes by $O(1)$. For lower semicontinuity, Lemma D.4 gives $D(\kappa) = \sup_{t>0} g_\kappa(t)$, an index set not depending on κ. For fixed $t > 0$, $\kappa \mapsto I_\kappa(t) = \sup_\theta\{\theta t - \kappa\varphi(\theta)\}$ is convex and finite, hence continuous. So D is a supremum of continuous functions, and with monotonicity it is left-continuous. The lower-semicontinuity argument applies to $c \mapsto D(cK, a)$ for any K satisfying Assumption A6, while monotonicity in c uses the token-discarding argument and hence needs all tokens to share one NTP, which holds in both instances of Assumption A5. (ii) Apply Theorem 5.3(b) with $K'(1) = \kappa\varphi'(1)$. (iii) The count vectors are i.i.d. across blocks under the prior, with the binomial laws of Lemma B.2; apply Theorem D.1 to the reduced data. □

Proposition E.1 (χ^2 regime)**.** *Under Assumption A5, if $K'(2) \le a/2 + K(2)/2$, then $D(K, a) = a/2 + K(2)/2$, where $K(2)L = (1 + o(1)) \sum_{t\in block} \log(1 + \chi^2(F_{P_t}\|\mathrm{Unif}))$. In the small-freedom instance of Corollary 5.4, this gives $D_{\text{full}}(\kappa, a) = a/2 + \kappa/4$ for $\kappa \le \kappa_0(a) \equiv a/(2\varphi'(2) - \frac{1}{2})$.*

With little evidence per segment, a second-moment calculation sets the boundary. Each watermarked block contributes χ^2-information $n^{K(2)}$, buying exactly $K(2)/2$ beyond the i.i.d. boundary. For Gaussian blocks this is the Donoho–Jin line $p = 1/2 + r$.

Proof of Proposition E.1. Let $p \in [K'(2),\ a/2 + K(2)/2]$, a nonempty interval by assumption. Since $K'(\theta) \le p$ on $[0, 2]$, $\inf_{\theta\in[0,2]}\{K(\theta) + (2-\theta)p\} = K(2)$, so $E_2 = a - 2p + K(2) \ge 0$, and Lemma D.3 gives $p \le D$. Thus $D \ge a/2 + K(2)/2$. Conversely, take any $p \in (a/2 + K(2)/2,\ a)$. Then $E_1 < 0$ and $E_2 \le a - 2p + K(2) < 0$. Also $p > K'(2) \ge t_1$, so $I_1^+(p) = I(p) - p$, and $I(p) \ge 2p - K(2) > a$ (take $\theta = 2$), hence $E_3 < 0$. By the proof of Theorem D.1(a) the problem is undetectable, so $D \le p$. Finally $\Lambda_n(2) = \sum_{t\in\text{block}} \log(1 + \chi^2(F_{P_t}\|\mathrm{Unif}))$. For the small-freedom instance, $K(2) = \kappa\varphi(2) = \kappa/2$ and $K'(2) = \kappa\varphi'(2)$, so the condition is $\kappa\varphi'(2) \le a/2 + \kappa/4$, i.e., $\kappa \le \kappa_0$. □

Proposition E.2 (Unknown alignment and segment length)**.** *In either instance of Assumption A5, let segments be arbitrary intervals of length $\ge \ell_n$, with ℓ_n unknown. Running ψ^* on k shifted grids for each block length in a geometric grid, with a Bonferroni correction, detects whenever $p < D\big((k/(k+1))^2 K, a\big)$, hence every $p < D(K, a)$ for k large. The lower bound of Theorem 5.3(a) still applies.*

κ	0.2	0.5	1	1.5	2	3	4
D_{full} (oracle)	0.400	0.475	0.576	0.634	0.669	0.698	0.700
D_{cnt} (key-only)	0.391	0.452	0.546	0.607	0.647	0.689	0.700
order-blind	0.350	0.350	0.350	0.350	0.350	0.350	0.350

Table 2: Largest detectable p in the small-freedom limit at $a = 0.7$ ($\kappa_0 = 0.588$, saturation at $\kappa^* = 3.353$). Achievability and impossibility exponents were computed independently and agree to grid precision.

The idea is that some shifted grid fits a full block inside each segment, losing at most a $(k/(k+1))^2$ fraction of its evidence, and $O(k \log n)$ grids cost only a polylogarithmic multiplicity.

Proof of Proposition E.2. Some grid length ℓ' satisfies $\ell'(1+1/k) \le \ell_n \le \ell'(1+1/k)^2$. Block starts of the k shifted grids are spaced ℓ'/k apart, so each segment contains a full block of some grid. Counting full signal blocks over all grids gives $\gtrsim |S|/\ell'$, so one grid contains $\gtrsim |S|/(k\ell')$ of them. Its signal-block fraction is $n^{-p+o(1)}$. In either instance all tokens share the NTP P_n, so a full signal block of length ℓ' has cumulant $c_n \Lambda_n$, with $c_n = \ell'/\ell_n \in [c_0, 1]$, $c_0 = (k/(k+1))^2$. The ratio c_n need not converge, so we argue along subsequences. Every subsequence has a further one with $c_n \to c \ge c_0$, along which the block cumulants converge to cK. Since all tokens share the NTP P_n, discarding tokens reduces a c-problem to a c_0-problem, so $D(cK, a) \ge D(c_0 K, a) > p$ and ψ^* has risk tending to 0 along it. Hence $R_n \to 0$. Lower semicontinuity of $c \mapsto D(cK, a)$ at $c = 1$ gives the final claim. The remaining blocks of that grid may be partly watermarked rather than null. This only helps, because $f_P(y) = \sum_w y^{1/P_w - 1}$ is increasing for every NTP P, so a watermarked pivot is larger than a uniform one in likelihood-ratio order, and T_k (a sum of the increasing functions $\log f_{P_t}$) is then stochastically larger on a partly watermarked block than on a null block. The same holds for the block counts, whose summands $1\{Y_t > 1 - u\Delta_n\}$ are increasing in Y_t. The counts entering $\mathrm{HC}_N^{\mathrm{dy}}$ can therefore only increase, and the lower bounds on them used in the proof of Theorem D.1(b) still hold. There are $O(k^2 \log n)$ tests. Replacing $(NL)^{-1}$ by $(NL^3)^{-1}$ in ψ^* makes the Bonferroni sum of type I errors vanish without changing any exponent. □

Adaptivity of the key-only test.

Remark E.3 (What the key-only test needs)**.** At a fixed u, the count test uses the threshold $1 - u\Delta_n$, and hence the scale Δ_n, as well as the block length ℓ_n, but it never uses the next-token probabilities themselves. The scale and length can be removed. Fix $\eta > 0$, scan thresholds $s \in \{(1+\eta)^{-j} : 1 \le j \le \log_{1+\eta} n\}$ together with the block-length grid of Proposition E.2, and apply a Bonferroni correction over these $O(k^2 \eta^{-1} \log^2 n)$ tests (including the shifts), using the level $(NL^3)^{-1}$ of Proposition E.2. This does not change any exponent. Some grid threshold equals $u'\Delta_n$ with u' within a factor $1 + \eta$ of any target u. The map $u \mapsto D(K_u, a)$ is lower semicontinuous (for fixed $t > 0$ the function g of Lemma D.4 depends continuously on u, and D is a supremum of such functions), so $\{u : D(K_u, a) > p\}$ is open. Hence, for η small enough, the scan detects every $p < D_{\mathrm{cnt}}(\kappa, a)$ (a routine compactness argument handles the dependence of u' on n, using that the $o(1)$ in Lemma B.2 is locally uniform in u, by its explicit formulas). With a factor-2 grid, as in the coarse adaptivity argument, it would instead be guaranteed only the worst case, over the position of u^* within a grid cell, of the better of the two neighboring grid points.

F Experiments: E2 figure and E3–E5

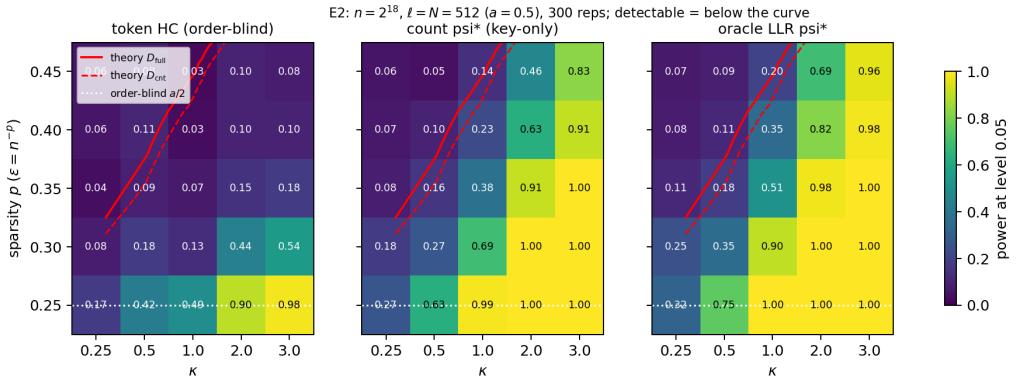


Figure 4: E2: empirical power over (p, κ) with the theoretical boundaries (red: D_{full} solid, D_{cnt} dashed; white dotted: order-blind $a/2$).

E3: unknown alignment and length. Segments of length 100 are placed at random ($n = 5 \times 10^4$, $\Delta = 0.1$; Figure 5).

- Misalignment costs a single grid substantial power, 0.53 against 0.84 for the aligned oracle at 1600 survivors.
- Four shifted grids recover most of the loss (0.68), and a fully length-adaptive multiscale scan is as good (0.71), as Proposition E.2 predicts.

E4: heterogeneous NTPs, unknown Δ. This setting lies outside the sharp theory, which needs the same NTPs in every block. With i.i.d. random NTPs this holds only approximately, for typical blocks. Each token is near-deterministic ($\Delta_t = 0.01$) with probability 0.6, and otherwise $\Delta_t \sim \text{Unif}(0.1, 0.5)$, with the residual mass split over $\{1, 2, 5, 20\}$ tokens (mean $\Delta_t \approx 0.13$). The key-only test scans four thresholds (Figure 5).

- With 800 survivors, its power rises from 0.08 to 0.25, 0.85 and 1.00 as ℓ goes from 1 to 25, 100 and 400, while order-blind tests stay between 0.15 and 0.40.
- Under i.i.d. survival ($\ell = 1$) the sum test is better (0.29). The combined test gives 0.17 there and 1.00 at $\ell = 400$, which supports it as the practical default.

E5: short documents. The design of E1 at an essay-like length, $n = 2000$, for $\ell \in \{1, 25, 100\}$ and two-point NTPs with $\Delta = 0.1$ (0.031 nats per token) and $\Delta = 0.4$ (0.17 nats per token), with 2000 replications per setting (Figure 6). The prediction is that block tests gain over order-blind ones once a segment's evidence $\ell \cdot \text{KL}$ is comparable to $\log N = \log(n/\ell)$.

- Order-blind tests are flat in ℓ, as in E1 (the sum test has power 0.33, 0.33 and 0.28 at 200 survivors and $\Delta = 0.1$).
- At $\Delta = 0.1$, a sentence carries $0.8 \ll \log 80$ nats, and at $\ell = 25$ the block tests are no better than the sum test (0.69 vs. 0.75 at 400 survivors). A paragraph carries $3.1 \approx \log 20$ nats, and at $\ell = 100$ the oracle block test gains (0.56 against 0.28 at 200 survivors, and 0.39 for the key-only test).

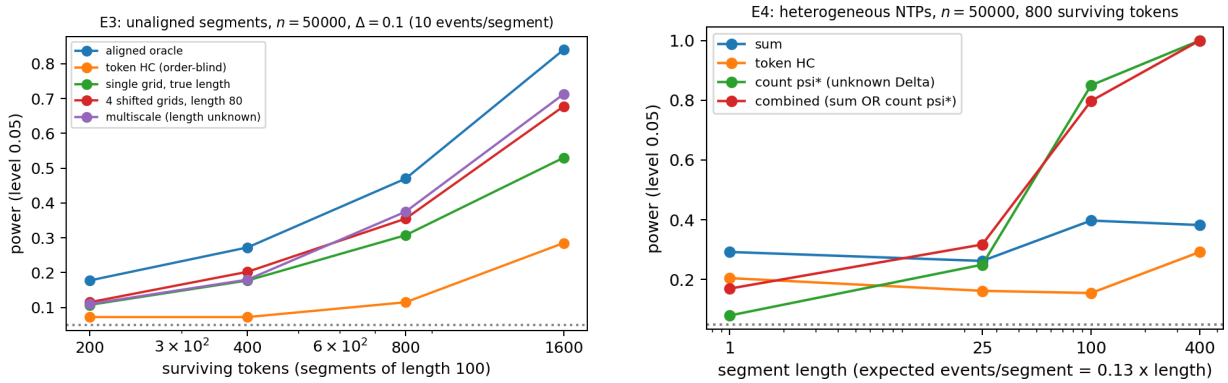


Figure 5: Left, E3: unaligned segments of unknown length. Right, E4: heterogeneous NTPs with unknown Δ_t.

- At $\Delta = 0.4$, a sentence carries $4.4 \approx \log 80$ nats, and the gain is large. With 100 survivors (four sentences) the oracle and key-only block tests have power 0.89 and 0.62, against 0.29 for the sum test. One surviving paragraph is detected with power 0.98 by the key-only test, against 0.28.
- The combined test loses at most 0.17 in power to the better of the sum and key-only tests (worst under i.i.d. survival).

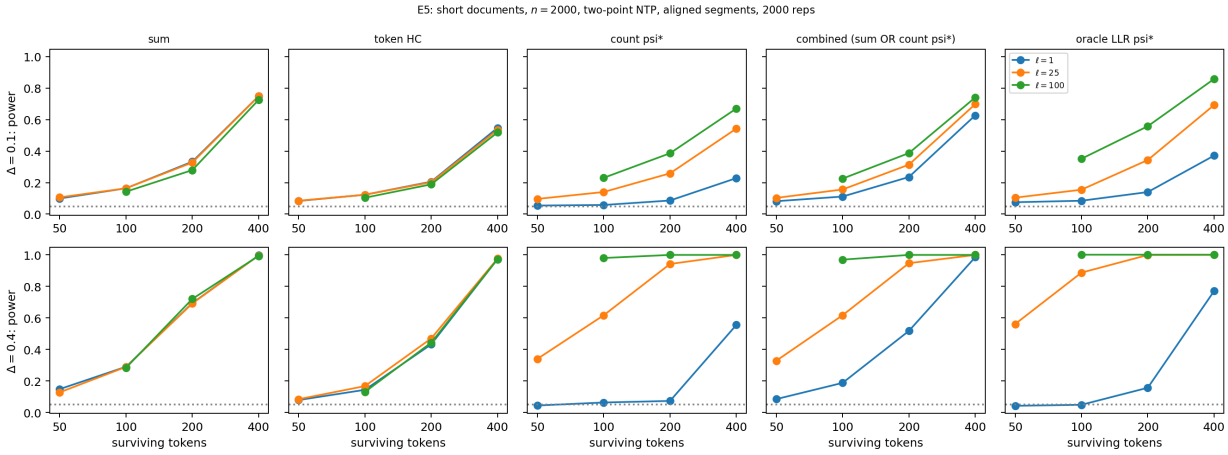


Figure 6: E5: power at $n = 2000$ against the number of surviving tokens; top row $\Delta = 0.1$, bottom row $\Delta = 0.4$.

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